

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

PAPER 2 Theory

4003/2

2 hours

SPECIMEN PAPER

Additional materials:
 Calculator (Optional)
 Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of **four** compulsory sections, **A, B, C and D**.

The Periodic Table is provided on Page 15.

FOR EXAMINER'S USE

Section A

SECTION B

Section C

Section D

TOTAL

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Section A

For
Examiner's
Use

- 1 (a) Fig 1.1 shows the human respiratory system.

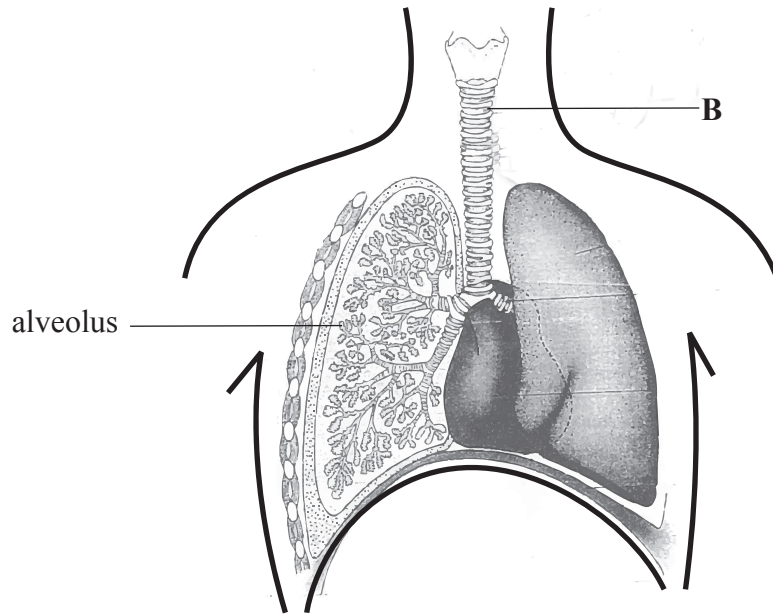


Fig. 1.1

- (i) Name the part labelled B.

[1]

- (ii) State any **one** feature of the named part that enables it to clean the air.

[1]

- (b) State the function of the alveolus.

[1]

- (c) (i) Name the inspired gas used in respiration.

[1]

- (ii) Describe a positive chemical test for the gas stated in (c)(i).

[2]

- (d) State the gas whose percentage concentration remains the same during breathing in and breathing out.

[1]

- 2 (a) Fig. 2.1 shows an agent of sugar bean pollination.

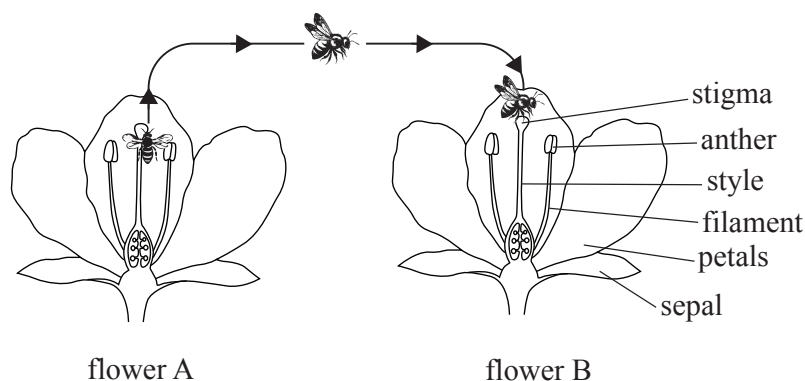


Fig. 2.1

- (i) Define the term *pollination*.

[1]

- (ii) State the type of pollination shown in **Fig. 2.1**.

[1]

- (b) Contact insecticides are chemicals which kill insects.
The chemicals can remain active for more than 14 days after application.

- (i) Explain the effect of using a contact insecticide on a crop of sugar beans at flowering stage.

[2]

- (ii) Describe how fertilisation takes place in a sugar bean flower.

[2]

- (iii) Describe the change which occurs to the ovary after fertilisation.

[1]

- 3 (a) A learner washed and placed an iron hoe in a damp shade. After three days, a brown substance had formed on the hoe.

- (i) Identify the process responsible for the formation of the brown substance.

[1]

- (ii) State **two** conditions necessary for the process identified in (a)(i) to occur.

[2]

- (iii) Suggest an improvement that can be done to the shade in order to reduce the occurrence of the process identified in (a)(i).

[1]

- (b) A manufacturer decides to supply hoes that are made up of galvanised iron.

- (i) Explain how the galvanized hoe would resist the process identified in (a)(i).

[2]

- (ii) Suggest **one** disadvantage of galvanising the hoe.

[1]

4 A dilute acid and a metal react to produce a salt and a gas.

- (a) (i) Name the gas produced.

[1]

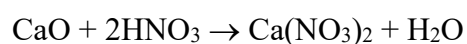
- (ii) State the type of bonding in the gas.

[1]

- (iii) Identify any **one** industrial use of the gas.

[1]

- (b) Calcium oxide and nitric acid react to form a salt and water as shown in the equation.



- (i) Name the salt produced.

[1]

- (ii) Calculate the mass of one mole of the salt produced.

[3]

- 5 (a) (i) Define the term *energy*.

_____ [1]

- (ii) Describe how sound is produced.

_____ [1]

- (b) Fig. 5.1 shows two cellphones in two jars.

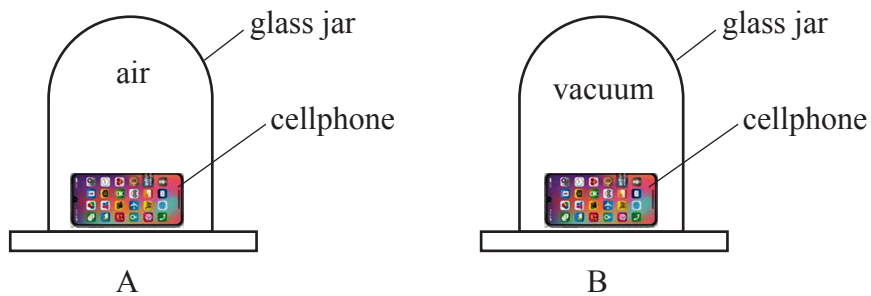


Fig. 5.1

- (i) Describe how the sound of the cellphone in **A** reaches the owner when it rings.

_____ [2]

- (ii) Explain why the sound of the cellphone in **B** **cannot** be heard when it rings.

_____ [1]

- (c) **Fig. 5.2** shows an incomplete light ray diagram.

Complete the diagram in **Fig. 5.2** to show the formation of a shadow.

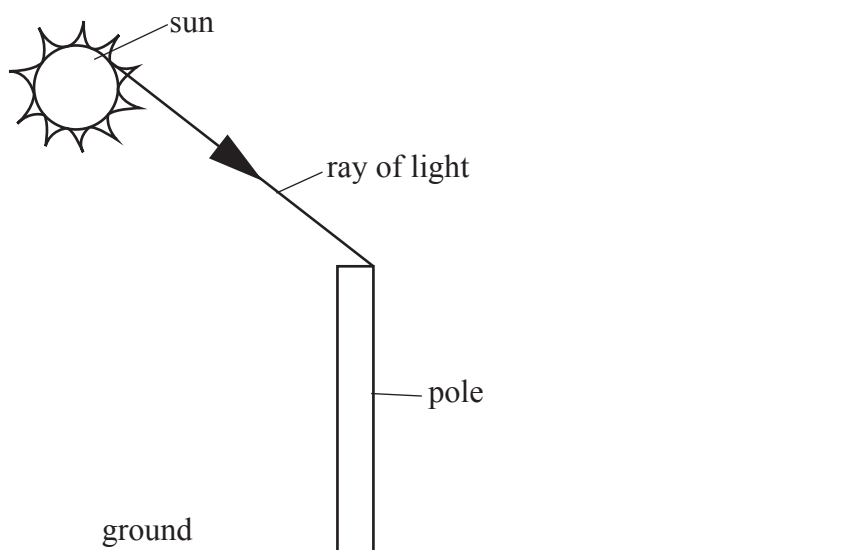


Fig. 5.2

- 6 (a) (i) Define the term *telecommunication*.

_____ [1]

- (iii) Long ago, kudu horns and drums were used for communication in Zimbabwe.

Identify any **two** modern ways used for communication in Zimbabwe.

1. _____

2. _____

[2]

- (b) Explain why a cellphone can be described as a transceiver.

[1]

- (c) Fig. 6.1 shows a communication medium.

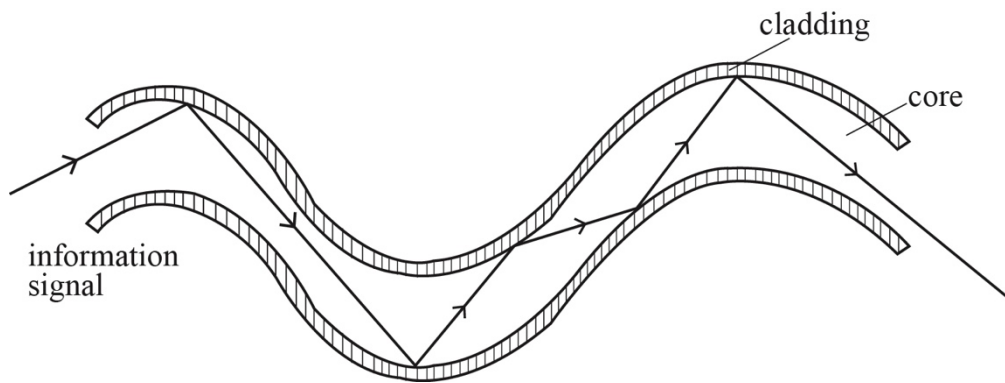


Fig. 6.1

- (i) Identify the medium.

[1]

- (ii) Outline how messages are transmitted through the medium named in (c)(i).

[2]

Section B

Answer any **two** questions in the answer booklet provided.

- 7 (a) (i) Write the word equation for photosynthesis. [2]
- (ii) State any **one** use of each of the products of photosynthesis. [2]
- (b) (i) Describe any **two** leaf adaptations for photosynthesis. [2]
- (ii) Explain the adaptations described in (c)(i). [2]
- (c) Name any **two** food storage organs in a plant. [2]

- 8 **Fig. 8.1** shows how the rhythm method is related to the menstrual cycle.

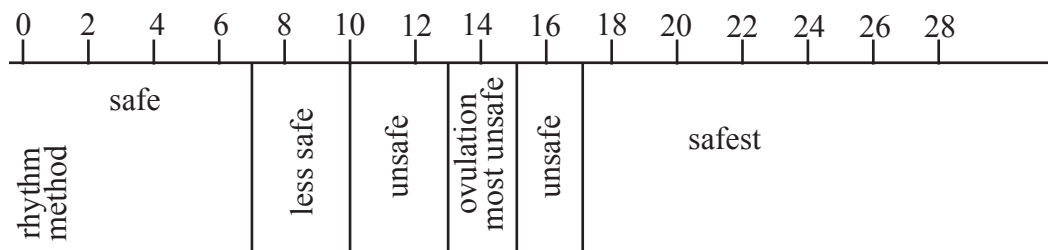


Fig. 8.1

- (a) (i) Name any **one** hormone which controls the menstrual cycle. [1]
- (ii) Explain why the ovulation period is the most unsafe period. [2]
- (iii) Explain why the period between days 10 and 13 is unsafe. [2]
- (b) (i) State any **two** advantages of using the rhythm method. [2]
- (ii) Give any **one** other natural method of contraception. [1]
- (c) The use of a condom falls under barrier methods.
- (i) Describe how the use of the condom prevents fertilisation. [1]
- (ii) State **one** other advantage of using the condom apart from preventing fertilisation. [1]

- 9 (a) Most low lying areas in Zimbabwe are still prone to malaria. The malaria vectors thrive in these areas.

Identify the malaria vector.

[1]

- (b) Fig. 9.1 shows the stages in the life cycle of the malaria vector.

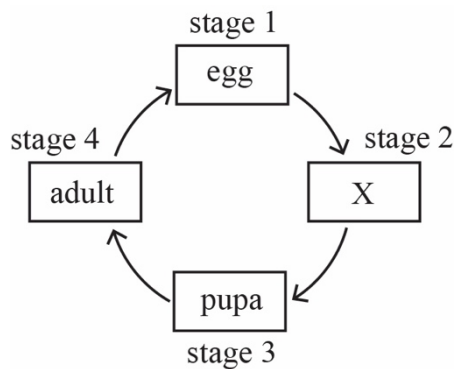


Fig. 9.1

- (i) Name the stage of the cycle represented by X. [1]
- (ii) Suggest any **two** methods that can be used to break the cycle at
1. stage 1,
 2. stage 4.
- [4]
- (iii) State any **two** signs or symptoms of malaria. [2]
- (c) Explain effects of malaria to a community. [2]

Section C

Answer any **two** questions in the answer booklet provided.

- 10** **Fig. 10.1** shows the apparatus used to separate two miscible liquids, A (boiling point 120 °C) and B (boiling point 60 °C).

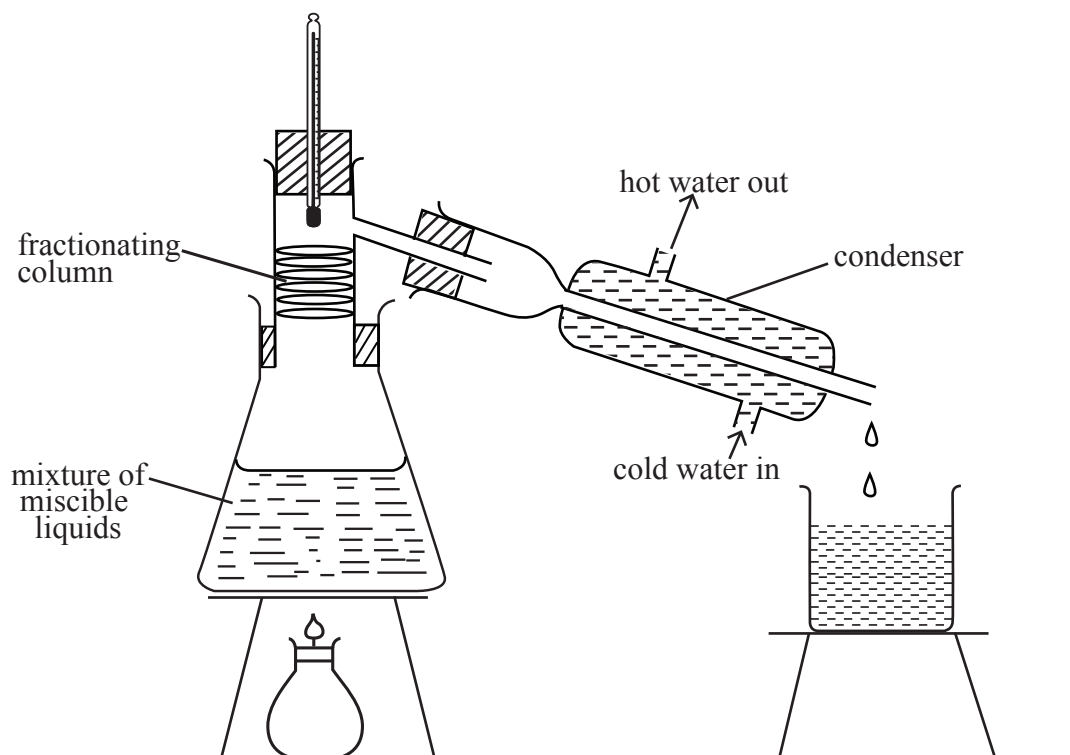


Fig. 10.1

- (a) Identify the method of separation shown in **Fig. 10.1**. [1]
- (b) (i) Describe and explain how the two liquids are separated. [4]
- (ii) Explain why cold water gets into the condenser through the lower opening. [2]
- (c) Explain any **one** industrial application of the method shown in **Fig. 10.1**. [3]

- 11 **Table 11.1** shows electronic configurations of elements represented by letters **A** to **F**.

Table 11.1

Element	Electronic configuration
A	2.1
B	2.8.2
C	2.8.7
D	2.6
E	2.8.1
F	2.8

- (a) State and explain using letters, **A** to **F**,
- (i) **two** elements which are in the same Group of the Periodic Table, [2]
 - (ii) any **one** element which forms a negative ion, [2]
 - (iii) an element which is unreactive. [2]
- (b) (i) Describe how chemical bonding occurs between elements **C** and **E**. [3]
- (ii) State the proton number of the element **B**. [1]

- 12 **Fig. 12.1** shows displayed formula of an organic compound.

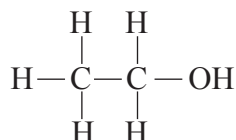


Fig. 12.1

- (a) (i) Name the organic compound. [1]
- (ii) State the homologous series to which the compound belongs. [1]
- (iii) Give any **one** use of the compound. [1]
- (b) A hydrocarbon contains 86% carbon and 14% hydrogen by mass.
- (i) Calculate the empirical formula of the hydrocarbon. [3]
 - (ii) Determine the molecular formula of the hydrocarbon, given that the relative molecular mass is 42. [2]
 - (iii) State the **two** products of the complete combustion of the hydrocarbon. [2]

Section D

Answer any **two** questions in the answer booklet provided.

- 13 (a) (i) Name the instrument used to measure force. [1]
- (ii) State the Standard International (SI) unit of force. [1]
- (b) (i) Define the term *moment of a force*. [1]
- (ii) **Fig. 13.1** shows a spanner being used to undo a rusty nut from the wheel of a scotch cart.

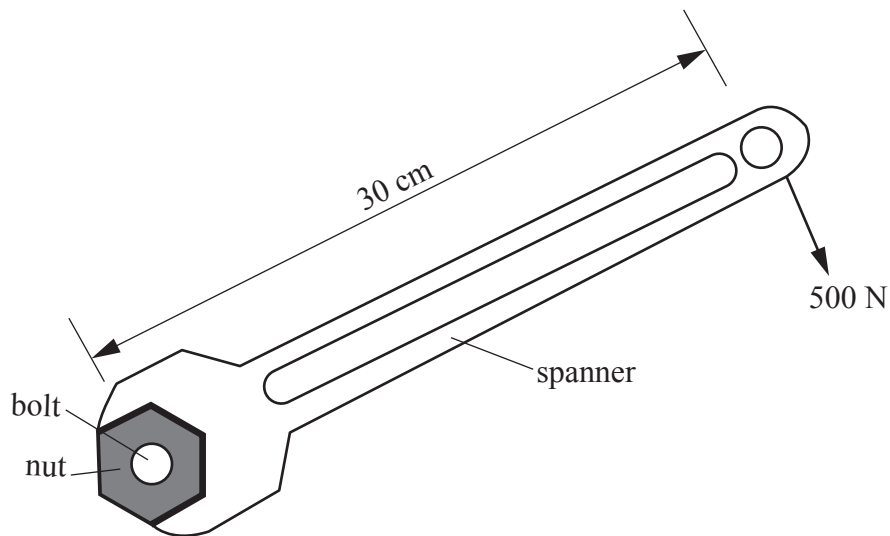


Fig. 13.1

- Calculate the moment generated. [3]
- (iii) Explain the effect of the rust between the nut and the bolt on the amount of effort applied. [2]
- (d) Describe **two** ways of making the removal of the nut easier. [2]
- 14 (a) List any **two** uses of electricity in the home. [2]
- (b) (i) State any **two** safety features in a three pin plug. [2]
- (ii) Explain how any **one** of the safety features stated in (b)(i) works. [1]

- (c) **Table 14.1** shows Zimbabwe Electricity Supply Authority (ZESA) electricity tariffs for a certain period of time.

Table 14.1

electricity tariffs	
energy/kWh	cost per unit
1 – 50	4 cents
51 – 100	7 cents
101 – 200	9 cents
201 – 300	12 cents
301 – 400	14 cents

- (i) Calculate the cost of buying 250 kWh of energy. [3]
- (ii) Give any **two** ways of saving electricity in the home. [2]

- 15** (a) Give any **three** components of a robot. [3]
- (b) Describe any **two** uses of fixed robots. [2]
- (c) Explain why a **NOT** gate is also called an inverter. [2]
- (d) Outline the stages in robot production. [3]

DATA SHEET

The Periodic Table of the Elements

Group																							
I	II											III	IV	V	VI	VII	O						
												¹ H Hydrogen											⁴ He Helium
7 Li Lithium	9 Be Beryllium											11 B Boron	12 C Carbon	14 N Nitrogen	16 O Oxygen	19 F Fluorine	20 Ne Neon						
3	4											5	6	7	8	9	10						
23 Na Sodium	24 Mg Magnesium											27 Al Aluminum	28 Si Silicon	31 P Phosphorus	32 S Sulphur	35.5 Cl Chlorine	40 Ar Argon						
11	12											13	14	15	16	17	18						
39 K Potassium	40 Ca Calcium	45 Sc Scandium	46 Ti Titanium	51 V Vanadium	52 Cr Chromium	55 Mn Manganese	56 Fe Iron	59 Co Cobalt	59 Ni Nickel	64 Cu Copper	65 Zn Zinc	70 Ga Gallium	73 Ge Germanium	75 As Arsenic	79 Se Selenium	80 Br Bromine	84 Kr Krypton						
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36						
85 Rb Rubidium	88 Sr Strontium	89 Y Yttrium	91 Zr Zirconium	93 Nb Niobium	96 Mo Molybdenum	98 Tc Technetium	101 Ru Ruthenium	103 Rh Rhodium	106 Pd Palladium	108 Ag Silver	112 Cd Cadmium	115 In Indium	119 Sn Tin	122 Sb Antimony	128 Te Tellurium	127 I Iodine	131 Xe Xenon						
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54						
133 Cs Caesium	137 Ba Barium	139 La Lanthanum	178 Hf Hafnium	181 Ta Tantalum	184 W Tungsten	186 Re Rhenium	190 Os Osmium	192 Ir Iridium	195 Pt Platinum	197 Au Gold	201 Hg Mercury	204 Tl Thallium	207 Pb Lead	209 Bi Bismuth	209 Po Polonium	209 At Astatine	210 Rn Radon						
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86						
Fr Francium	226 Ra Radium	227 Ac Actinium																					
87	88	89																					
*58-71 Lanthanoid series †90-103 Actinoid series																							
		140 Ce Cerium	141 Pr Praseodymium	144 Nd Neodymium	150 Pm Promethium	152 Eu Europium	157 Gd Gadolinium	159 Tb Terbium	162 Dy Dysprosium	165 Ho Holmium	167 Er Erbium	169 Tm Thulium	173 Yb Ytterbium	175 Lu Lutetium									
		232 Th Thorium	238 Pa Protactinium	238 U Uranium	238 Np Neptunium	244 Pu Plutonium	244 Am Americium	244 Cm Curium	244 Bk Berkelium	247 Cf Californium	251 Es Einsteinium	252 Fm Fermium	257 Md Mendelevium	259 No Nobelium	261 Lr Lawrencium								
		90	91	92	93	94	95	96	97	98	99	100	101	102	103								
Key		a = relative atomic mass X = atomic symbol b = proton (atomic) Number																					
		a																					
		X																					
		b																					

*58-71 Lanthanoid series
+ 90-103 Actinoid series

a

a = relative atomic mass

X

X = atomic symbol

b

b = proton (atomic) Number

140 Ce Cerium	141 Pr Praseodymium	144 Nd Neodymium	147 Pm Promethium	150 Sm Samarium	152 Eu Europium	157 Gd Gadolinium	159 Tb Terbium	162 Dy Dysprosium	165 Ho Holmium	167 Er Erbium	169 Tm Thulium	173 Yb Ytterbium	175 Lu Lutetium
232 Th Thorium	232 Pa Protactinium	238 U Uranium	238 Np Neptunium	244 Pu Plutonium	244 Am Americium	244 Cm Curium	244 Bk Berkelium	244 Cf Californium	244 Es Einsteinium	244 Fm Fermium	244 Md Mendelevium	244 No Nobelium	244 Lr Lawrencium

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)